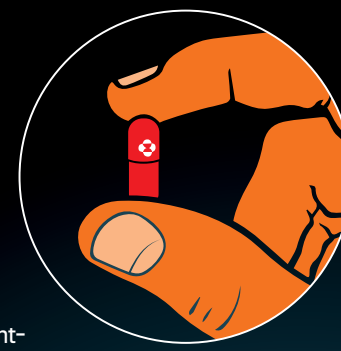
NEW LINES OF TREATMENT



The phones don't stop ringing at Delhi's Manipal Hospital. Worried kin of Covid patients are calling constantly, a majority insisting their loved ones be given the latest wonder treatment—monoclonal antibodies (maBs) cocktail.

We have been there before. If it was the demand for hydroxychloroquine and plasma therapy in the first wave, the second wave saw remdesivir, ivermectin, steroids such as dexamethasone and tocilizumab as the drugs du jour. This time, it's maBs.

The antibody cocktail had proved highly effective in preventing severe disease and death during the second wave. Dr Naresh Trehan, chairman of Medanta-The Medicity, the first hospital to introduce the treatment in the country, had at the time said that it held "great promise in reducing hospitalisation rate and could even be administered to children". MaBs are lab-prepared antibodies that mimic antibodies in the human body when injected for a faster immune response against Covid. Administered intravenously or subcutaneously in a 2-4-hour procedure in a hospital setting, the antibodies target the spike protein of the virus to stop it from replicating within the body.

In May 2021, India's Central Drugs Standards Control Organisation granted emergency use authorisation to Regeneron, and its partner Roche, for their Casirivimab and Imdevimab antibody cocktail, known by the brand name REGEN-COV. A single dose costs between Rs 59,750 and Rs 65,000 in India. Global data of 4,567 patients published by Roche in March 2021 showed that the antibody cocktail reduced hospitalisation or death by a fifth and shortened the duration of symptoms or hospitalisation by four days. The study was conducted at a time when the Delta variant was the dominant Covid strain in the world. "Antibody cocktails significantly reduced the progression of the disease," says Dr Rahul Pandit, director of ICU at Fortis Hospitals, Mumbai. "But it is important that they are administered at an early stage—usually 3-5 days after onset of symptoms."

Many experts, however, are sceptical about its efficacy against Omicron. On December 16, Regeneron issued a statement saying its antibody therapy had "diminished potency versus Omicron but was active against the predominant Delta variant". A pre-print in BioRxiv studied the efficacy of nine antibody cocktails against Omicron and found it to be significantly resistant.

Pharmaceutical firms have already begun testing new antibodies for use against Omicron. Early research holds promise for two in particular—otrovimab, developed by San Francisco-based Vir Biotechnology and London-based GSK, and DXP-604, developed by Beijing-based BeiGene and Singlomics. Neither is presently available in India.

ANTIBODY COCKTAILS CUT DOWN THE RISK OF HOSPITAL STAY DURING DELTA, BUT THEIR EFFICACY AGAINST OMICRON IS UNCERTAIN

There is, however, a new drug already in the market that private hospitals and physicians are free to prescribe to patients. Molnupiravir, an antiviral pill by Merck, received EUA in India last month by the DCGI (Drugs Controller General of India). Over a dozen labs, including Cipla, Dr Reddy's and Sun Pharma, will produce the Merck pill, as it is popularly known, in India. With a starting price of Rs 1,399 for a five-day course, studies show it reduces hospitalisation and death in early-stage infections by half. ICMR chief Dr Balram Bhargava, however, cautions against the pill's side-effects and it hasn't been added to the national Covid treatment guidelines. These side-effects can trigger foetal and genetic defects as well as cartilage damage. The pill is also dangerous for pregnant women and needs to be taken alongside three months of contraception, according to Dr Bhargava.

A second antiviral pill, Nirmatrelvir by Pfizer, is also due for approval in India soon. Results from Phase II and III clinical trials show that it reduced the risk of hospitalisation or death by 89 per cent. "It takes some time before one can start prescribing new drugs. You have to be cautious and keep in mind the various side-effects. Overall, Omicron has had mild symptoms, infecting mostly the upper respiratory tract. Medication at this point should be given to those at higher risk or those showing signs of progression to severe disease," says Dr S. Kumar, medical director, LNJP Hospital, Delhi. That's some wise advice against an unpredictable virus.